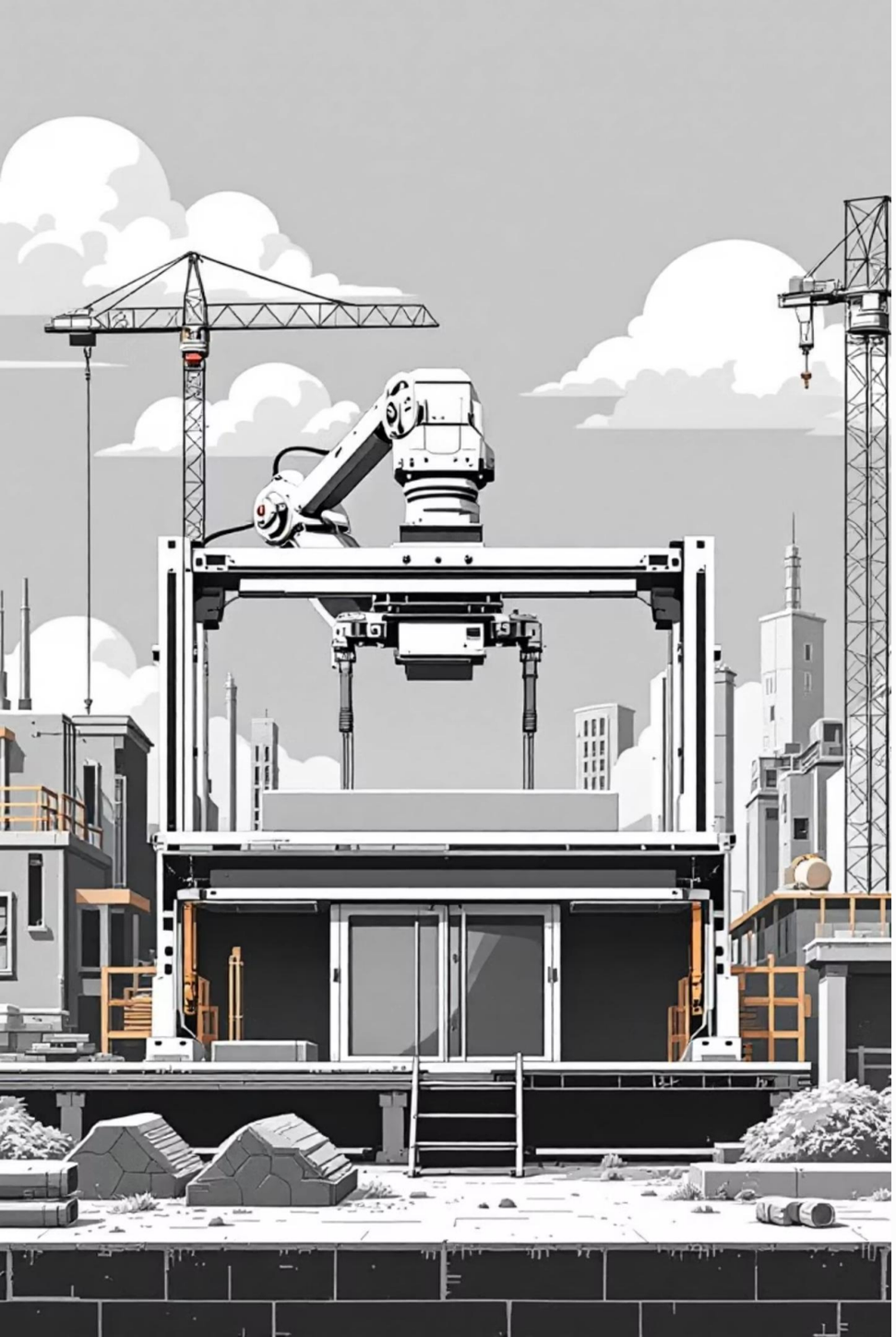


Intelligent 3D Concrete Printing

Layer Deformation Monitoring and Shrinkage Prediction Using Machine Learning

A research initiative by Jeetu Kumar, Sagar Hiremath, and Dr. Sowmya Kamath S from the Department of Information Technology, National Institute of Technology Karnataka, Surathkal

The Challenge: Transforming Construction

Traditional Construction

- Time-consuming and labour-intensive
- Heavy reliance on formwork
- Limited architectural freedom
- High material waste

3D Concrete Printing Potential

- 50–60% cost reduction
- Eliminates formwork requirements
- Complex geometries enabled
- Minimised material waste

What is 3D Concrete Printing?

An additive manufacturing process that fabricates concrete structures layer-by-layer through controlled extrusion of specially formulated cementitious material, eliminating formwork and enabling unprecedented design freedom.

01

Digital Model

Design specifications converted to machine instructions.

02

Material Extrusion

Cementitious material deposited layer-by-layer through nozzle.

03

Automated Construction

Robotic systems build complex structures without formwork.

04

Quality Finishing

Integrated trowels smooth surfaces for paint-ready finish.



Critical Challenges in 3DCP

Layer Stability

Maximum printable layers before lower layers deform under weight of subsequent layers.

Process Control

Precise calibration of nozzle diameter, extrusion pressure, print speed, and layer height required.

Durability Issues

Shrinkage and crack formation threaten long-term structural integrity and interlayer bonding.

Research Objectives

1. Maximum Layer Stability

Determine the maximum number of layers printable before lower-layer deformation exceeds acceptable limits.

2. Real-time ML Control

Develop machine learning system to detect local deformations and dynamically adjust print speed.

3. Shrinkage and Crack Prediction

Predict shrinkage and crack formation using process, material, and environmental data

Data Foundation: Input Features

Material Composition

Cement, water, silica fume, fly ash, aggregates, fibres, and admixtures (kg/m³).

Process Parameters

Print speed, nozzle area, layer height, interlayer time, extrusion pressure.

Environmental Conditions

Ambient temperature and relative humidity affecting setting time and drying shrinkage.

Rheological Properties

Yield stress, plastic viscosity, buildability index, and deformation rate indicators.

Predictive Outputs: What We Measure

Compressive Strength

Load-bearing capacity (MPa)

Flexural Strength

Bending resistance (MPa)

Tensile Strength

Interlayer bonding quality (MPa)

Layer Deformation

Vertical settlement (mm)

Shrinkage Strain

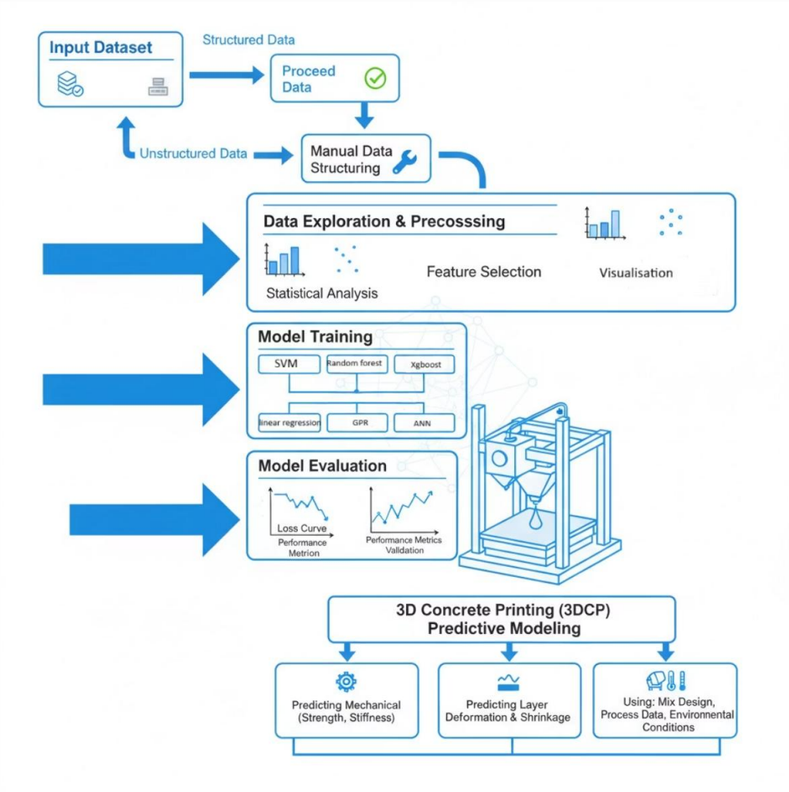
Dimensional instability (%)

Crack Propagation

Width and rate (mm/s)

Machine Learning Approach

Leveraging advanced ML algorithms including SVM, XGBoost, random forests, and neural networks to model complex non-linear relationships between material composition, process parameters, and structural performance.



Predictive Models

SVM, XGBoost, ensemble methods for deformation and shrinkage prediction.

Real-time Monitoring

Adaptive control adjusting print speed based on detected deformations.

Performance Validation

High correlation coefficients demonstrating model reliability and accuracy

Project Deliverables and Impact

- ## Curated Datasets

  Comprehensive layer deformation, material properties, and environmental condition records from NITK 3DCP Laboratory.

- ## Predictive ML Models

  Validated models for layer deformation and shrinkage/crack initiation prediction.

- ## Decision Support System

  Analysis and visualizations guiding optimal printing parameters and mix design decisions.

- ## Industry Transformation

  Enhanced reliability, structural performance, and efficiency enabling wider 3DCP adoption in modern construction.

Thank you